ChatGPT

Probability of Range Selection

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If a population has a mean of 125 with a standard deviation of 25, what is the probability of randomly selecting a value in the range of 75-175?

Assuming the population is **normally distributed** with mean $\mu = 125$ and standard deviation $\sigma = 25$:

Compute z-scores:

$$z_1 = \frac{75 - 125}{25} = \frac{-50}{25} = -2, \qquad z_2 = \frac{175 - 125}{25} = \frac{50}{25} = 2.$$

So the probability is

$$P(75 \le X \le 175) = P(-2 \le Z \le 2) = \Phi(2) - \Phi(-2) = 0.9772498681 - 0.0227501319 = 0.954$$

Answer: **≈ 0.9545**, or **95.45%** (about 95.4%).

(If you don't assume normality, Chebyshev's theorem only guarantees at least $1 - 1/2^2 = 75\%$ will lie within 2 standard deviations.)

Two different normal distributions could have the same mean but different standard deviation.

Question 4 options:
True
False

Answer: True

Two normal distributions can have the **same mean** but **different standard deviations**.

- The **mean** decides the center of the distribution.
- The **standard deviation** decides how spread out the data is.

So both curves can be centered at the same point, but one can be wider (larger standard deviation) and one narrower (smaller standard deviation).

Therefore, the correct answer is **True**.

What is the probability of finding P(Z ≤ 0.72)?

Enter your answer using 5 digits after the decimal.

We need to find:

$$P(Z \leq 0.72)$$

Using the standard normal table (or calculator):

$$P(Z \leq 0.72) = 0.76424$$

Final Answer (5 decimal places):

$$\boxed{0.76424}$$

What is the probability of finding P(Z ≥ -0.85)?

Enter your answer using 5 digits after the decimal.

We are asked for:

$$P(Z \geq -0.85)$$

Step 1: Use the symmetry of the standard normal distribution:

$$P(Z \geq -0.85) = 1 - P(Z < -0.85) = 1 - \Phi(-0.85)$$

Step 2: Use the property $\Phi(-z) = 1 - \Phi(z)$:

$$\Phi(-0.85) = 1 - \Phi(0.85)$$

Step 3: Look up $\Phi(0.85)$ in the standard normal table:

$$\Phi(0.85) \approx 0.80234$$

Step 4: Compute $P(Z \geq -0.85)$:

$$P(Z \geq -0.85) = 1 - (1 - 0.80234) = 0.80234$$

✅ **Final Answer (5 digits):**

$$\boxed{0.80234}$$

P(Z≥−0.85)\=1−P(Z<−0.85)\=1−Φ(−0.85)
explain me why did you do 1 - p(z)?

Ah, I see! Let me explain clearly.

We are asked:

$$P(Z \geq -0.85)$$

By definition:

$$P(Z \geq a) = 1 - P(Z < a)$$

This is because the **total probability under the normal curve is 1**. The area under the curve is split into two parts at any point a:

- The area to the **left** of a: $P(Z < a)$
- The area to the **right** of a: $P(Z \geq a)$

Since the total area is 1:

$$P(Z \geq a) + P(Z < a) = 1 \implies P(Z \geq a) = 1 - P(Z < a)$$

In our case, $a = -0.85$, so:

$$P(Z \geq -0.85) = 1 - P(Z < -0.85)$$

We then used the property of the standard normal table that:

$$P(Z < -z) = 1 - P(Z < z)$$

to find the value easily from a standard table.

If you want, I can draw a **quick diagram** showing why this "1 – area" trick works—it makes it very visual. Do you want me to?

but why did you do it ? you could also write 1 - 0.80234 = value

Ah, now I understand your confusion! Let me clarify.

We **cannot directly do `1 - 0.80234`** here, because **0.80234 is already** $P(Z \geq -0.85)$ in the standard normal table calculation we used.

Here's the sequence I used:

1. Standard normal tables usually give $P(Z \leq z)$, the area to the left of z.
2. We want $P(Z \geq -0.85)$, which is the area **to the right** of -0.85.
3. By the total area rule:

$$P(Z \geq -0.85) = 1 - P(Z < -0.85)$$

4. But $P(Z < -0.85)$ is **to the left of -0.85**, which tables don't always show directly. So we use symmetry:

$$P(Z < -0.85) = 1 - P(Z < 0.85) = 1 - 0.80234 = 0.19766$$

5. Now plug in:

$$P(Z \geq -0.85) = 1 - P(Z < -0.85) = 1 - 0.19766 = 0.80234$$

✅ That's why the final answer is **0.80234**.

So we **didn't just subtract 0.80234 from 1**, because 0.80234 comes from $P(Z < 0.85)$, not $P(Z \geq -0.85)$ directly.

If you like, I can **draw a simple diagram of the normal curve** showing all these areas—it makes this super obvious visually. Do you want me to do that?

What is the probability of finding P(Z ≤ 1.53)?

Enter your answer using 5 digits after the decimal.

We are asked for:

$$P(Z \leq 1.53)$$

Step 1: Look up the standard normal table for $Z = 1.53$.

$$\Phi(1.53) \approx 0.93699$$

✅ Final Answer (5 digits):

$$\boxed{0.93699}$$

What is the probability of finding P(-0.12 ≤ Z ≤ 1.23)?

Enter your answer using 5 digits after the decimal.

Question 8 options:

We are asked for:

$$P(-0.12 \leq Z \leq 1.23)$$

Step 1: Use the cumulative probability formula

$$P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$$

Here, $a = -0.12$ and $b = 1.23$.

Step 2: Look up the standard normal table

$$P(Z \leq 1.23) = 0.89065$$

$$P(Z \leq -0.12) = 0.45224$$

Step 3: Subtract the two probabilities

$$P(-0.12 \leq Z \leq 1.23) = 0.89065 - 0.45224 = 0.43841$$

✅ Final Answer (5 digits):

$$\boxed{0.43841}$$

P(Z≤−0.12)\=0.45224
i think you did 1 - (1-0.54776)

Ah, yes! Let me clarify that part carefully.

We want $P(Z \leq -0.12)$.

Step 1: Use the symmetry of the standard normal curve

$$P(Z \leq -z) = 1 - P(Z \leq z)$$

Here, $z = 0.12$, so:

$$P(Z \leq -0.12) = 1 - P(Z \leq 0.12)$$

Step 2: Look up $P(Z \leq 0.12)$

$$P(Z \leq 0.12) \approx 0.54776$$

Step 3: Compute $P(Z \leq -0.12)$

$$P(Z \leq -0.12) = 1 - 0.54776 = 0.45224$$

✅ That's exactly why the value is **0.45224**.

So yes, what you wrote:

$$1 - (1 - 0.54776)$$

is essentially the same as this symmetry formula, just written a bit differently.

If you want, I can **draw a small visual** showing why this symmetry works—it makes it much easier to remember. Do you want me to do that?